

Advanced (Habanero)

- Q1.** What is the GCF of 12 and 18?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 9
- Q2.** What is the LCM of 9 and 12?
(A) 18
(B) 24
(C) 27
(D) 36
(E) 48
- Q3.** Find the GCF of 24 and 30 using prime factorization.
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12
- Q4.** What is the LCM of 8 and 15?
(A) 30
(B) 40
(C) 60
(D) 120
(E) 150
- Q5.** Find the GCF of 50 and 75.
(A) 5
(B) 10
- (C) 25
(D) 50
(E) 75
- Q6.** A bookshelf has 12 shelves, and each shelf can hold 8 books. If the bookshelf is currently empty, what is the LCM of the number of shelves and the number of books each shelf can hold?
(A) 12
(B) 24
(C) 48
(D) 96
(E) 192
- Q7.** A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, what is the GCF of the number of pieces of candy and the number of friends?
(A) 1
(B) 2
(C) 4
(D) 8
(E) 16
- Q8.** What is the LCM of 6 and 9?
(A) 12
(B) 18
(C) 24
(D) 27
(E) 36

Open Ended Questions (FRQ)

- Q9.** A bakery is making a special batch of cookies for a holiday sale. They need to package 48 cookies into boxes that hold 8 cookies each, and 24 cookies into boxes that hold 4 cookies each. What is the GCF and LCM of the number of cookies in each type of box?
- Q10.** A water tank can hold 120 gallons of water. If 8 gallons of water are added to the tank every hour, and 4 gallons of water are removed from the tank every 2 hours, what is the LCM of the number of gallons added and removed from the tank in a 24-hour period?

Chef's Tips

- Emphasize the importance of prime factorization in finding GCF and LCM.
- Use real-world examples to illustrate the application of GCF and LCM.
- Encourage students to check their work by verifying that the GCF divides both numbers and the LCM is a multiple of both numbers.